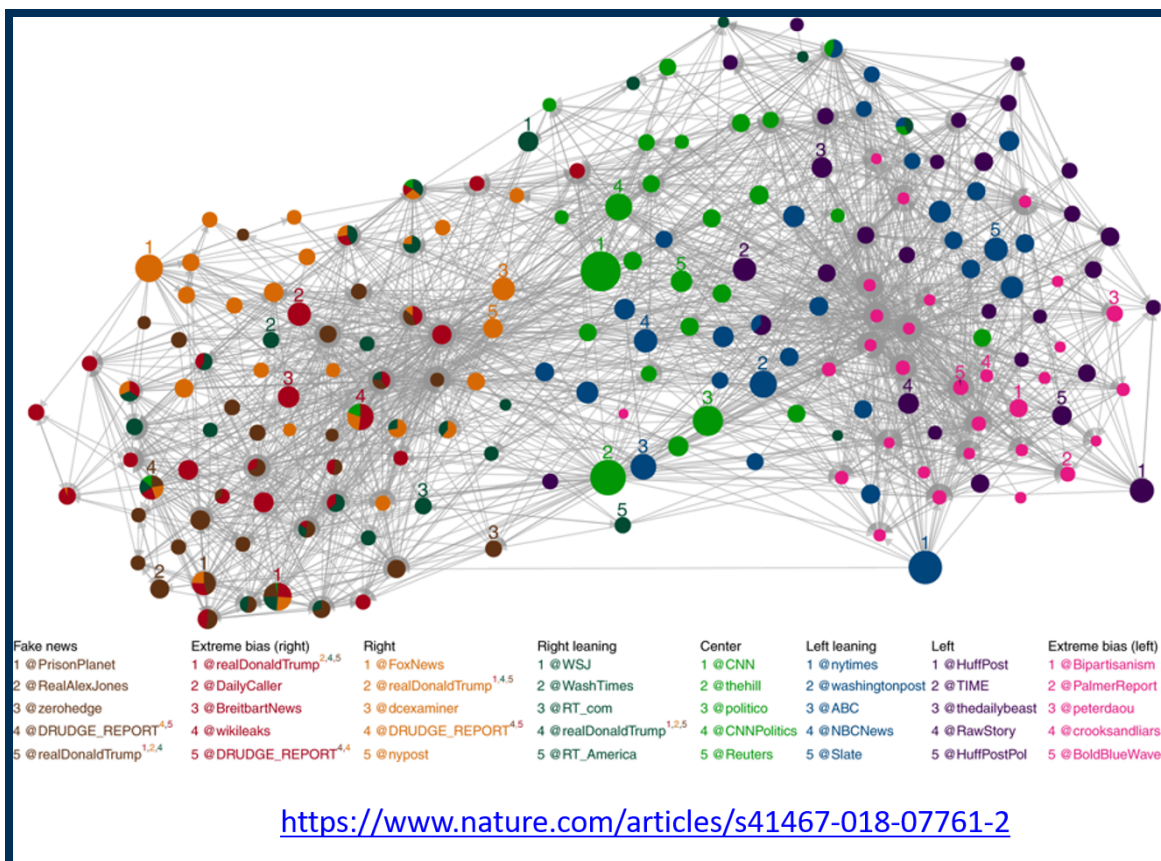


L1: Influence and Cascade Phenomena

The dynamic processes that take place on a network are often not physical. For example, ideas, opinions, and other social trends and hypes can also spread through networks – especially over online social networks.

We will study such influence or “information contagion” phenomena in the context of mostly Facebook and Twitter.



For example, the image above comes from a recent study focusing on the effect of misinformation (*known as “fake news”*) on Twitter in the 2016 US Presidential Elections. (Image Source: [Bovet, A., Makse, H.A. Influence of fake news in Twitter during the 2016 US presidential election. Nat Commun 10, 7 \(2019\)](https://www.nature.com/articles/s41467-018-07761-2) <https://www.nature.com/articles/s41467-018-07761-2>)

The study used network science to identify the most influential spreaders of fake news as well as traditional news.

An important but still open research question is whether it is possible to develop algorithms that can identify influential spreaders of false information in real-time and block them.

Source Links

[Influence of fake news in Twitter during the 2016 US presidential election](https://www.nature.com/articles/s41467-018-07761-2) 
(<https://www.nature.com/articles/s41467-018-07761-2>)